

UNIVERSITY RESULT MANAGEMENT SYSTEM

PHASE 1: PROJECT INITIATION

1.1 Introduction

1.2 Project Goal

1.3 Stakeholders

1.4 Planning

1.5 Scope

PHASE 2: PROJECT PLANNING & SCHEDULING

2.1 Project Evaluation & Review Technique (PERT)

2.1.1 Draw the Network Diagram

2.1.2 Time Estimation

- Optimistic Time
- Most Likely Time
- Pessimistic Time

PHASE 3: COST ANALYSIS

3.1 Cost Calculation using Critical Path Method (CPM)

PHASE 4: MANAGEMENT CONCLUSION

4.1 Timeline Chart based on Time Calculation

4.2 Software Cost Estimation

PHASE 5: FINAL SUBMISSION

5.1 Conclusion

5.2 Submission of the Report

PHASE 1: PROJECT INITIATION

1.1 Introduction

In modern educational institutions, managing student academic records and results manually is a complex, time-consuming, and error-prone process. With the increasing number of students, subjects, and examinations, universities require a fast, accurate, and reliable system to handle result processing efficiently.

The **University Result Management System** is a software-based application designed to automate the complete process of result generation, storage, and publication. This system helps in recording marks, calculating grades and CGPA, generating mark sheets, and displaying results online securely.

This project focuses on **project planning, scheduling, and cost estimation** using project management tools like **PERT (Project Evaluation and Review Technique)** and **CPM (Critical Path Method)**. These techniques help estimate project duration, identify critical activities, and calculate the total project cost.

The system improves efficiency, reduces manual errors, ensures data security, and provides quick access to academic results for students, faculty, and administrators.

1.2 Project Goal

The main goal of the University Result Management System is to develop an **automated, reliable, and user-friendly system** for managing university results.

Objectives:

- To automate the result processing system
- To reduce paperwork and manual workload
- To ensure accurate mark calculation

- To generate result reports automatically
- To allow students to access results online
- To maintain a centralized secure database
- To improve transparency and efficiency

1.3 Stakeholders

Stakeholders are individuals or groups who are directly or indirectly involved in the project.

Primary Stakeholders:

1. **University Administration** – Oversees academic management
2. **Examination Department** – Manages result processing
3. **Faculty Members** – Enter and verify student marks
4. **Students** – View and download results
5. **System Administrator** – Maintains the software system
6. **Project Development Team** – Designs and develops the system

1.4 Planning

Project planning includes organizing tasks, scheduling activities, allocating resources, estimating time and cost, and managing risks.

The overall project plan consists of:

- Requirement gathering
- System analysis
- Feasibility study
- System design
- Database design
- Coding
- Testing
- Deployment

To ensure timely completion, **PERT and CPM techniques** are applied for proper planning and scheduling.

1.5 Scope

In Scope:

- Student login
- Faculty login
- Mark entry
- Grade calculation
- Result generation
- Online result publication
- Report generation

Out of Scope:

- Online exams
- Attendance tracking
- Fee management

PHASE 2: PROJECT PLANNING & SCHEDULING

2.1 Project Evaluation and Review Technique (PERT)

PERT is a statistical tool used to analyze project activities and estimate the total project time. It uses three-time estimates:

- **Optimistic Time (To)** – Best possible time
- **Most Likely Time (Tm)** – Normal time
- **Pessimistic Time (Tp)** – Worst-case time

PERT Formula:

$$TE = \frac{To + 4Tm + Tp}{6}$$

2.1.1 Draw the Network

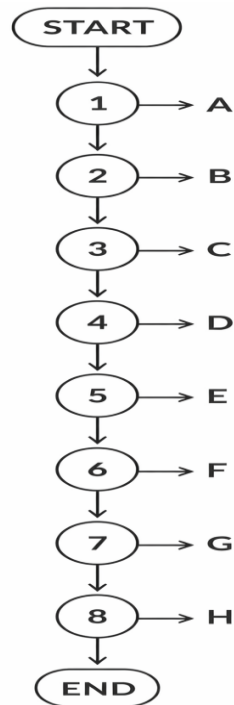
Activity List

Activity	Description
A	Requirement Collection
B	Requirement Analysis
C	Feasibility Study
D	System Design
E	Database Design
F	Coding
G	Testing
H	Deployment

Immediate Predecessor Table

Activity	Predecessor
A	-
B	A
C	B
D	C
E	D
F	E
G	F
H	G

Network Diagram



2.1.2 Arrive with Time Required for the Project

Time Estimation Table (Days)

Activity	To	Tm	Tp
A	3	5	7
B	4	6	8
C	2	4	6
D	5	7	10
E	4	6	9
F	8	12	16
G	4	6	9

H	2	3	5
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Expected Time Calculation

$$TE = \frac{To + 4Tm + Tp}{6}$$

Activity	TE
A	5
B	6
C	4
D	7
E	6
F	12
G	6
H	3

Total Project Duration

$$5+6+4+7+6+12+6+3=49 \text{ Days}$$

Total Project Duration = 49 Days

PHASE 3: COST ANALYSIS

3.1 Cost Calculation using CPM

CPM identifies the **critical path**, which is the longest duration path.

Critical Path:

A → B → C → D → E → F → G → H

Manpower Requirement

ROLE	COUNT
Project Manager	1
System Analyst	1
Software Developer	2
Tester	1
Database Engineer	1

Total Engineers = 6

Cost Estimation Table

Role	Cost/Day (₹)	Days	Total (₹)
Project Manager	1500	49	73,500
System Analyst	1200	49	58,800
Developers (2)	1000*2	49	98,000
Tester	800	20	16,000
Database Engineer	1000	15	15,000

Total Project Cost

$$73,500 + 58,800 + 98,000 + 16,000 + 15,000 = ₹2,61,300$$

Total Estimated Cost $\approx$ ₹2.6 Lakhs

PHASE 4: MANAGEMENT CONCLUSION

4.1 Timeline Chart (Activity Duration)

Activity	Duration (Days)
A	5
B	6
C	4
D	7
E	6
F	12

G	6
H	3

Total = 49 Days

This timeline ensures structured workflow, smooth execution, and on-time completion.

4.2 Software Cost Estimation

The estimated project cost of **₹2.6 Lakhs** covers:

- Development cost
- Testing cost
- System deployment
- Maintenance support

This ensures high-quality and reliable software delivery.

PHASE 5: FINAL SUBMISSION

5.1 Conclusion

The University Result Management System is a modern solution for automating academic result processing. The application improves efficiency, reduces errors, saves time, and provides secure online access to results.

Using PERT and CPM, proper project scheduling and cost estimation are achieved. The project is planned for completion within **49 days** at a cost of **₹2.6 Lakhs**.

This system enhances transparency, accuracy, and academic management efficiency.

5.2 Submission of the Report

The complete project documentation containing planning, scheduling, network diagrams, time calculations, cost analysis, and conclusions is prepared and submitted for academic evaluation.